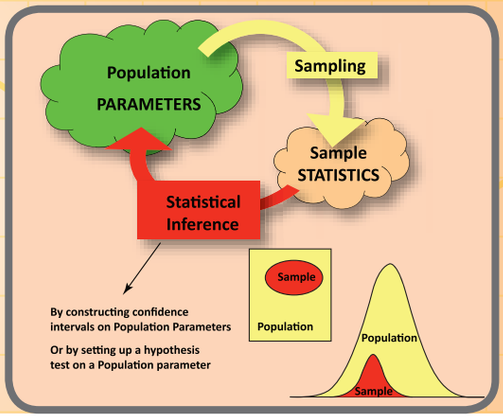
CHAPTER

1 TESTS OF SIGNIFICANCE – BASIC CONCEPTS AND LARGE SAMPLE TESTS





Jerzy Neyman

Jerzy Neyman (1894-1981) was born into a Polish family in Russia. He is one of the Principal architects of Modern Statistics. He developed the idea of confidence interval estimation during 1937. He had also contributed to other branches of Statistics, which include Design of Experiments, Theory of Sampling and Contagious Distributions. He established the Department of Statistics in University of California at Berkeley, which is one of the preeminent centres for statistical research worldwide.



Egon Sharpe Pearson

Egon Sharpe Pearson (1885-1980) was the son of Prof. Karl Pearson. He was the Editor of *Biometrika*, which is still one of the premier journals in Statistics. He was instrumental in publishing the two volumes of *Biometrika Tables for Statisticians*, which has been a significant contribution to the world of Statistical Data Analysis till the invention of modern computing facilities.

Neyman and Pearson worked together about a decade from 1928 to 1938 and developed the theory of testing statistical hypotheses. *Neyman-Pearson Fundamental Lemma* is a milestone work, which forms the basis for the present theory of testing statistical hypotheses. In spite of severe criticisms for their theory, in those days, by the leading authorities especially Prof.R.A.Fisher, their theory survived and is currently in use.

"Statistics is the servant to all sciences" – *Jerzy Neyman*

LEARNING OBJECTIVES

The students will be able to

- understand the purpose of hypothesis testing;
- define parameter and statistic;
- understand sampling distribution of statistic;
- define standard error;
- understand different types of hypotheses;
- determine type I and type II errors in hypotheses testing problems;
- understand level of significance, critical region and critical values;
- categorize one-sided and two-sided tests;
- understand the procedure for tests of hypotheses based on large samples; and
- solve the problems of testing hypotheses concerning mean(s) and proportion(s) based on large samples.

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